

ActInf Textbook Group ~ Cohort 1 ~ Meeting 14 (Welcome back, starting of

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 14 (Welcome back, starting of Chapters 6-10) | 54:07

okay we begin

welcome everyone it is

the resumption of

our cohort

update the video description it's the

resumption of cohort one of the active

inference textbook group

it's our 14 okay could someone mute

Giuseppe or feel free to say whatever

you want to say

well uh can you hear me yes yes okay so

sorry but I'm completely new to this

interface of a gather and uh encoder so

I I'm not sure I understand cohort one

and cohort 2 here is that so because

cohort one was the wasn't it the

previous one like the the one that yes

the start last year

well it's still going on

from May till July

we did chapters one through five

okay now we are in this cohort so if

those who did not participate in this

part like respectfully you know thanks

for the question but everyone in general

respectfully let's try to have people

who are in the cohort participate in it

everyone's welcome to join and of course

re-watch but contributions in the

questions and the chat are probably best

so that we can respect the people who

are continuing on

this is our 14th meeting and now we're

heading into the second five chapters of

the book

okay does any other person want to give

oh yeah go ahead

hold on so the the the the current one

that is going on right now is the cohort

one and not cohort two we're all

learning in different stages so the

previous hour was the cohort that is

focusing on chapters one through five

okay stage they're in and now some of us

a different set of people some are

joining both so that's why we made the

meetings convenient was so that we could

like for many of us who want to do it

both we can join or there's but

this is a separate set of people

I see I see yes so okay so cohort true

was was first and then I mean right in

the this meeting in this meeting right

now cohort 2 was first yes and now

which started correct before we warmed

up the chapters one through five

okay so I

read it because I I subscribed for

cohort too so excellent okay good thank

you so much see you next time thanks for

setting up all this yeah this thing

okay bye bye yes awesome

okay

anyone else want to add anything

we don't need to do a coda overview

I would say people who are watching this

recording can look at the live recording

if they want a code to update that we

just went through

for the cohort twos live meetings

but does anyone

want to give any um

introductory thoughts otherwise there's

a few ways I think we can go for this session
but we're here just to rejoin our activities and Orient towards chapters six or ten
we can scan through the chapters we can have any suggestions or discussions around ways to approach them I don't know
uh Rohan and then anyone else I was wondering when are we gonna get to the projects part or discussions about the projects
great question let's look at the chapters and see where that first off it can happen however you want with contacting people and
um like we can make a sub page like I see exists here this looks really awesome
so people can request a sub page they can mention it that contact them now or however
so it's up to you for a project that is under your control
it can be developed in the open here or however
you can also participate in many ongoing projects like active block fronts which also
um which might all share this are get coin Grant was approved yesterday
so if people are familiar with this or want to become familiar this is a way to like provide or receive support possibly
for their learning in blockburns and

contributing to that open source package

[Music]

um

does that address what you were thinking

were you thinking about these project

ideas or were you thinking more about

like the sort of build your own model

approach that will be starting in

chapter six

I know about uh project ideas in the

quota page

awesome yeah like these are

um

well they're they're different different

types of projects clearly more will be

added but they're of different

they're not necessary they might involve

building a generative model but they are

not

degenerative model itself or only

let's look at chapter six

because this is where the generative

model construction is going to come into

play and so again there's a lot of

projects that

you would only need to qualitatively

think about generative models for like

educational material production

is related to the generative model or

could be modeled as a generative model

of a learner

but you wouldn't necessarily have to

make a general model but in chapter six

um

it's a recipe for Designing active

inference models

it is going to rock sorry just real

quick uh I think Luke is here uh doesn't

have access against or access

okay okay

message in Discord or I mean messaging

gather your email address and I'll add

you

Okay so

sorry I didn't

oh okay you requested access all right

I'll add you that's enough then

um

in chapter six it's going to ask some

questions

that are uh driving the design of the

generative model

which system of remodeling

system of Interest

what is the appropriate form for the

generative model

reflecting the discrete or continuous

distinction that we looked at

and also all these other questions that

aren't exactly within the textbook like

temporal depth hierarchical nesting

which parameters are learned versus

fixed all these different directions

that the models actually have to be

elaborated on even before getting to the

data engineering and maybe even code

implementation

just from a structural perspective what

form is it

how to set up the generative model

is where some of those analytical

details begin being introduced

not necessarily even the program level

implementation or an executable

simulation but like

okay we have a preference Vector that's

is that is there two states are is there
some continuous preference distribution
whatever it is
what is the actual details of the
generative model
and then
the other side of the coin slash blanket
is the generative process
so what are the endogenous Dynamics
and the um causal mapping with action
for the generative process
and there's kind of cases where you have
more of a communication setting active
inference entities on both sides
you could also have more of like a
simple simple
partnering
like two um deterministic programs that
send messages to each other or it could
be like one more cognitive
more agent-like generative model and one
more
environmental non-living type
any thoughts or overviews or how would
people think about
how can we make this type of a recipe
design should we be all designing a
model or how do we even go about reading
a section of the book in a chapter that
begins with a
recipe provision Lyle
yeah so so I I spent a lot of time
building what I think would be classed
here as generative models and what I
find is that this what I I characterize
it as the step of framing it up framing
up is is often far trickier sometimes
much harder frankly than building a

model depending on what what the situation is

and in my experience I find that it all comes down to making sure we understand what is the question that we're trying to answer right it's sort of like because because every modeling process I'm sure everybody on this call knows is the process of deciding what to leave out right what to put in what to leave out and what level of representation then it all has everything that's in there if it's going to be a good model has to be necessary and necessary to answer a question or provide a particular Insight or so forth so that whole issue of framing out uh what you're going to build you know and in all those different the 10 000 questions yeah there's more than four you know ultimately it's many then it's all about that which often means understanding the audience if if the model builder is the only audience this is a much simpler if the audience is

um

uh you know some much larger group with different perspectives a lack of of homogeneity and insight then there almost has to be uh I'm gonna say almost a pre-modeling or framing process of of sort of of framing up what are the questions we care about and then we can talk about how how those translate into a model

so

that's just some general framing

thank you that's very

interesting it it

um makes me think about like this

audience question for the model like why

are we building the model why are we

shaping our regime of attention to build

this model

building can be an end of itself it can

be creative and expressive and fun and

all these other things and maybe later

there's a road that has some pragmatic

value too but also it can just be like

fun like you know what would a

generative model that could do this look

like and that can be very didactic

there's other times where you're

presenting information

oh I made a simulation of Twitter

discourse and dot dot dot

is always going to be presented in a way

that's not only just like the model

distilling down from the world but then

you're communicating it and then when

you're actually not just communicating

but if you're making an application that

does something and communication is

doing something as well then

the system of Interest as Lyle mentioned

is totally non-trivial and there's the

whole framing

in some of these early stages especially

because they can't always be changed

easily later

um run and then anyone else

uh yeah uh so I'm just curious when you

say who's the audience uh

is it that uh like isn't this part of

the active inference model itself that
okay so whatever makes sense in that
framework wouldn't that be the audience
here
thank you
as in it has to survive right your agent
has to survive so whatever generic model
helps it survive
would that not be the
model of choice over here and I'm just
curious I'm not sure that's awesome it's
awesome like way to frame it and it's
it's um maybe someone can share the um
free energy governance link
in the in the chat I just can't look it
up right now but but um
it puts this very nicely in in a recent
pretty short book dissertation about
like that imperative for organizations
and um yes in one sense it can be said
the right generative model or like the
audience for it is whatever its
survivability conditions are but and
within our textbook group The
survivability conditions are open
however we're also learning about or
even just gesturing towards
like
okay
not just we're gonna imagine a room with
a person who can take on a jacket or
take it off
um we can make that a didactic model and
the audience could be this textbook
group
or future cohorts of the textbook group
and the function could be just sort of
like a

neutral

Exemplar

active inference generative model but
then of course the considerations for
information presentation empirical data
analysis any kind of application
the system of interest is not just the
one person with the jacket on or off two
states room with a sign fluctuation
so that's a different situation for
model building so is that how should we
interpret this recipe

oh okay okay

but is there not like a journal
framework for what a good generative
model looks like is what I'm saying like
I I guess you could frame that
definitely so if you want to build pixel
level representations for example you
can use again and that's a pretty decent
one but maybe pixel level
representations are not a good
uh not good for the task at hand or
something like that so it should be able
we should it should fit a general
category right so this is what a good
set of models looks like for active
inference so that that sort of is
getting it

yeah that's a very fascinating framing
does anyone want to add something to
that

is I I totally hear you like on what is
there a is there a rubric like what are
we evaluating the generative model
and process

fit

because it's not like there's just

simply a best generative model you know
bigger or you know whatever no future
about it
except outside of a generative process
um context is the fitness
and then what is their rubric is our
gold standard is it like oh you're
dealing with audio data here's the file
format
that level of standardization certainly
not
however are there some
um practices or lenses that we could
bring in from people's experience with
evaluating model adequacy and relevancy
there probably are
Lyle
well yeah that really comes right back
to purpose right so a model is always
fit for purpose
and and you of course this also surfaces
the questions of model validation right
one of the things we haven't talked
about here is what's the right model how
do we frame up the right model what do
we choose to put in what leave out part
of that is purpose and and how would I
evaluate you know if I do what I'm
imagining a model will be how do I
evaluate its quality its validity well
what data is actually available that's
one thing you always have to ask at the
outset what data is available what data
can I use what's computationally
you know adequate and what parts of this
then uh when we kind of project
you know when we apply this model if
that's its role then does it fit for the

purpose does it provide you know good
you know adequate results and improve to
be you know whatever Behavior we're
trying to address

so

you can't you can't say if it's good if
you don't have a purpose so if your
purpose is just to explore model
building which is something I love to do
then

that's a pretty big there's a lot of
things that are fit for that purpose

right but if the purpose is to

um uh you know Pro do a prospective uh
analysis of a system and what actions
might be taken to intervene then that's
in a different thing

thanks a lot for sharing it I'm I'm
really

happy and I think it's relevant that
we're having this kind of a discussion
because we've gone through some of the
first chapters

and we will continue through chapter six

I think into having this conversation

but

does anybody else want to add something
about this more General topic of like
how are we even coming to talk about
what system or how are we framing the
system that we're modeling

that's in section 6.3

I um

I mean this is too broad to it's not
active in French specific but

whenever I'm trying to model something
the only questions that I mean it is
contextual to the purpose it entails a

purpose in an audience but all I want to know basically is what are the questions that need this model needs to like generate an answer for um what is the information that is like necessarily therefore kind of required to Model A system that would have such answers in it uh and maybe another way to say the same thing is that like what affordances would the model acquire to answer address whatever the some set of questions that are posed to the system so if you asked you know I don't know um like how frequently does a yellow colored car I don't know drive down this particular Road um then I wouldn't need to record like the number of bikes or their colors uh that are you know using the bike lane or buses or trucks large you know so and like this is um maybe outside the scope of what I mean by yellow how many you know cars out of all the cars how many are yellow or something that's right driving down this road or whatever um it's like not a even exactly a generative model but my point is that modeling that system there there is obvious data that's like completely

irrelevant right

um and so

uh

if you were going to build some sort of

simulation of any kind that's going to

be generative like

there's often a lot of data that's just

completely extraneous depending on what

questions you're going to ask of it

so

thanks

in six three

in the first paragraphs

they're gonna come out hard on the

interface concept of Markov blanket

this is how you'll do system modeling is

by finding interfaces that distinguish

the internal and external states of a

system

then

that

is critiqued or elaborated

because there's an infinite degrees of

freedom

in the core screenings and those are

modeler specific

and they suggest that that matters

especially if you want to be considering

embodied extended perspectives

where just descriptive

interfacing models won't give you the

whole answer there's a great paper could

a neuroscientist understand a

microprocessor

showing that even if you have total

access to the microprocessors firing

like an EEG and a generated set of

single and double lesions

a lot of the interpretations would be strongly misleading

um yet people want the wiring Circuit of the brain which is dynamic and also involves diffusive molecules

we have all 302 nematode neurons mapped but that also as an interface diagram is not even close

so if you want to describe the nematode like model building then it's a fit for purpose

however there might be very simple models that could be doing a lot better in this in nematode Behavior observation the form is discussed there are three main design choices first discrete and continuous time we had some awesome discussions in the previous months about like similarities and differences and where like in theory they're different and then in practice where they're different

the whole question of like discretizing continuous systems

um the second choice is between shallow models and hierarchical models where inference operates at multiple time scales

the third choice is the consideration only of the present kind of the variational free energy versus models with temporal depth

temporal depth consideration could be purely passive

and it could also include some elements of affordance which if we remember to like the B Matrix actions are influencing how States the world are changing through time so that is how active inference is unifying policy comparison by kind of embedding action into estimates of how things could evolve probabilistically those three design decisions are explored in slightly more detail in the subsections six five how to set up the generative model after specification then the specific variables have to be included okay the latent State temperature is going to be continuous the observations in my thermometer I'm going to take in as just integers and then there's going to be a person who is going to be in a binary state of like whether it's hot or cold or maybe I want hot neutral cold or maybe I want them to have a 10 point scale or I want them to have um a scale that's doing Dot that is in this stage of actually setting up the generative model and giving a little bit of like color to what had been just outlined it's going to be a nested model with this continuous variable and this kind

of a discrete variable

um

here we return to the Frog

example from earlier in the book

where there's like but it's kind of

being generalized a little bit it was

introduced with the whole Bayesian

equation walkthrough now we're returning

to it and thinking more about multimodal

or like even you could consider this as

Sensor Fusion

because there's some hidden cause

the jumping or the action of a frog

that is inducing several domains of

impact potentially in the generative

process

and then that leads to specific

observables from the perspective of the

cognitive entity

those could be again two communicative

cognitive entities or one and not or

however

so this is like okay

we're going to be modeling

forest fires

so forest fires they change the

temperature they change the

chemical composition they change the

vibrations in the ground or something

okay and then maybe there's three

sensors maybe you have three

thermometers

and then what about them are you gonna

average them are they going to be like

three parallel modeled Time series

that is where some more specification of

the generative model comes into play

keeping

this the active inference ontology in
mind

what are the affordances the Frog what
are the latent States

that are being changing through time
like sound

physical presence or Photon emission or
something and then what are the data
that would be oh the observations
[Music]

um

this box looks like it's about

different sensory modalities I'm sure
there's a lot of interesting

discussions we could bring in here

priors and empirical Behavior

complete class theorems

any statistical decision procedure

Behavior may be framed as Bayes optimal

under the right set of Prior beliefs

discussion of fixed and learned
variables

one can imagine the pros and the cons
generally of having fixed and learned
variables

like

computationally it's simpler to fix a
hard-coded variable however if there's a
setting where the model's adequacy
depends on being able to update that
maybe you want it learnable

however then you have to deal with these
questions about like the kinetics of
learning like instantly switch or slow
switch or Trend averaging so there's a
lot of discussions around learning how
is attention

regulated what is precision on learning

it's like it very much complexifies the
model
and there's many variables that you can
bring into a learnable setting
but when it's framed as just static
variables
and only like the hidden States changing
through time that's like the simplest
case
but more and more cognitive entities
probably are going to entail
considering learning Dynamics
setting up the generative process
the reason we postponed the design of
the generative process
so
a little slate of hands
entity first not the um
Niche first is that in many practical
applications discussed in this book we
simply assume the Dynamics of the
generative process are the same or very
similar to the generative model
we generally assume the agent's
generative model closely mimics the
process that generates its observation
that is clearly a simplifying case
because it doesn't address
what if
you're paying attention to predicting
with and the world is dependent on
height
there's so many
complexities of that which is really
that broader like what is being modeled
what matters question
but in some
way very upstream or very close to the

Stream

it is like the rabbit being put in the

Hat

like if the rat knows that there's a 5x5

maze

that's a huge amount of information

versus if it only has access to Pure

local information

and uh that that's just something to

like

I think for us to note as we look

through models

like are we modeling the generative

process oh so the seasons oscillate so

it is a sine wave

so then the generative model of the

agent is also going to be fitting a side

wave

well then

it kind of collapses to like a

stochastic regression

and maybe then it looks like the fit is

going to be very good

active inference modeling brings in

another complexity with action

influencing outcomes

that prevents some

um I guess fallacies of modeling but by

no means all it probably introduces many

others Eric

yeah uh can you hear me uh this is a

news sounds good um so this this might

be a time to to um to discuss the

question I posted and I apologize I

didn't put it in the right place the

other day

um I just figured out today that it goes

under this particular location but this

is about the question about

um

why modeling of a generative process is necessary

is a necessary step for building an active inference model so let me just read it

um yeah because it's a lot of words uh so you know we know that an active inference agent employs a generative model as this model of the external world and it uses its General model to perform inferences about perceptual input and to make decisions about actors so in general the external world is complex and unknowable to the uh to the agent the generative model may or may not be able to learn but if learning is supposed to uh if it is learning is opposed to put the generative model in a better alignment with the world's generative process whatever that may be so I can see how one might model the generative process

if one is building an artificial agent in a laboratory where you kind of have access you you control how the world behaves with respect to the agent and then you might want to study how well the agent design

um Works versus you know one versus another under control conditions but this is not the same as designing an agent to go out and cope with some real world environment where you really don't know what the generative process is so I think that the recipe for designing an active interest models would

include formally setting up the generative process only in the kind of a laboratory situation but not if you're really trying to build something out you know for the for the real world because you don't

know the general process you might create an artificial laboratory kind of situation by trying to build a model of how the world works and say well how does my agent work under that those assumptions or not but that you know that might set you up for putting the agent in the real world but in in fact you don't know the degenerative process if you're really trying to build something to to work in the real real world

That's my kind of my question that's a question

thank you Lyle

well I I love uh Eric I love that that framing because I think there's multiple different cases right so I think if you're modeling something uh that as you say goes in in the real world then you know perhaps it's a you know some sentient creature or being um then

then it could obviously there's the real world even if we knew everything about it is too complex to model right so so that's that's Fool's Gold you know to assume you know everything right

there and and so I don't I'm just gonna say yeah that's hard you know typically I guess we would say what are the parts

of the generative the real world
degenerative uh process that we're going
to
consider relevant and deal with that and
knowing that we're making a leap of
faith but there's another case I think
which is is more generalizable just to
come a complex adaptive systems that
operate in what I call the built world
all right so uh I'm I work with a bunch
of people who build digital twins they
build digital twins of
um human infrastructure
right so buildings and all this stuff
they're the generative process that
we're concerned with is largely knowable
right and largely represent
representable and so and and what's
interesting about that is there are a
bunch of blind spots and inadequacies in
what we know about that generative
process and the ability to do inference
about what the real state of that system
is then becomes the value of this
approach in that setting right so so I
think I I don't think there's a general
answer once again I think it really
comes down to purpose and what the
specific framing you're looking to do to
me this ability to say something about
what I I like that idea of decision
agents right decision agents that uh are
going to potentially take action or
infer actual States
is is much richer than what a lot of the
modeling historical modeling paradigms
have given us so I think that's really
interesting

thank you Lyle Ali

uh yeah and uh I also think that the distinction between uh General surprises and generative models

becomes significant uh if we want to simulate learning rather than just inference uh I mean that's the answer I got from Ryan Smith because I asked him almost the same question why would why do we need generative process in order to

simulate the agents and

um because you see for the generative process

uh

if if the generative model exactly Maps them to generative process we only need that capital D Matrix right but uh and that just when we're just trying to stimulate the inference side of the agent right but when the learning comes along then we need to separate uh that capital d uh from another Matrix probably

the case D which Maps into the generator process so yeah one of the situations that the distinction between two might be necessary is when we when those two process I mean learning and inference are taken as two separate phenomena to model

thanks

um one note is also there's work on symmetries in the generative process generative model like I think like the bi-directional paper with Matt Sims and blue

um discussing it and then

um

here will be a defense to Eric's

excellent question

um

The Entity first modeling approach

that's taken by this chapter

one can say I don't know whether I'm

going to a forest or a desert but I'm

going to have these sensors on my

designed interface

and I'm going to do extensive modeling

on interception and homeostasis

and if I can learn policies that balance

homeostasis

then I've survived

if I fail then

failure happens

so that doesn't mean that it works or

it's doing the right thing fit for

purpose any of those aspects but one of

the insights from the 4E and the

embodied side is like a vast amount of

our

what is modelable as cognitive

isn't necessarily

the declarative

I think therefore I am

possibly even symbolic type thinking

linguistic

there's a lot going on

and many of that a lot going on

can be simplified by not even knowing

about the external process maybe not

even the as modeled kidney knowing about

what the as modeled liver is doing

so this type of a focus was like we're

going to Define

our entity types

and the kinds of observations that they
can make the kinds of affordances that
they have or do empirically or that they
conceptually could do

that approach

um is consistent with embodied

biological systems and Persistence of
organizations and

so I'm wondering how that piece comes
into play with this discussion that's
very important about and also their
their

earlier discussion of the embedded and
extended cognition is that it kind of
blurs the line

with what is a generative model what is
a generative process

so

hence there being a lot of

thinking and attention

on what the system's modeling is

what are the known unknowns and the
unknown unknown is going to be

let's just see

um

a chapter ends

six seven simulating visualizing

analyzing and fitting data using active
inference

so

standard routines for active inference

that provide support for all functions

are available

#spm

and appendix C

step-by-step model is a great

look through this this is Matlab code

one amazing work that we could all

iterate on would be porting Matlab
into the ontology
into Python and or Julia with pi mdp or
active block friends
but there's a lot in SPM and some of its
neuroimaging specific some of it's very
variational base specific not all of it
is going to be state-of-the-art
or the most useful in every situation
however
I wonder if only offering certain
routines in Matlab
which is like a traditionally very
academic
software platform and potentially
non-free
um
can be a limiting factor so
those who feel like they want to learn
more or contribute this way
helping
structure some of the code
architecture so it cohorts coming
through
can be doing multiple language maybe
first we can just move some of these
routines into a folder
and then people can start to organize
those types of um aspects
summary
it's a four-step recipe
we walk through it from the
consideration of the system of Interest
broadly like the entity and the process
focusing in on the entity specifically
thinking structurally about what it is
going to be like doing slash composed of
specifying that to a few levels of

detail then doing the same for the
generative process and then once that is
specified
it's not quite there
for most people today but
where
it seems like they are laying out a path
towards and where we probably can and
will be after that level of setup
in some potentially graphical editor or
parameter file
then
as as SPM and Matlab have done in many
many papers the generative model
specification almost prepares the
working script
but there's still a lot of details and
and pieces to add in especially in a
modern environment and for all of these
like non-standard and novel cases that
people are wanting to consider but that
is once we have a notebook that goes
through the recipe
then the recipes can be forked and so on
and they can be written in different
languages Pi mdp block friends Ali
yeah and just one thing that I think is
not made explicit in this chapter
is that actually that first step which
system are we modeling especially uh the
identification of the marker blanket
part is uh basically at this point of
active inference research uh is more
like a like a hypothetical step because
um
I I also had a misapprehension that uh
we can somehow uh identify the markup
blankets from data but then as I went on

through my project and by Consulting
Maxwell and others
um actually this is the comment Maxwell
uh made about my proposal that he said
that identifying markup blanket from
data is still an outstanding issue that
is not resolved yet in the uh research
and active inference and one of the most
active areas areas of research they're
doing at versus lab is just focused on
this part of the modeling recipe I mean
how to how we can
identify automatically identify or
somehow learn the markup blanket from
just pure data points
awesome and that ties in with Dalton s's
work on the blanket index and like
continuous variables for dynamic
Assemblies of blanketing like it this is
gonna get really
nuanced and empirical
with the discovery and the thresholding
of Markov blankets and causal inference
in Real World Systems with many
different kinds of sensors and doing
structure learning on generative model
and process and like everything
empirically which is why as a
simplifying step

[Music]

um
the uh it actually really reminds me of
how Dave Snowden
situates the difference between systems
thinking and complexity thinking
which is systems often uses like a lot
of nodes and flow charts to represent
system elements and relationships and

like interfaces whereas in complexity in
the canvas especially
um framing
it's like more like
self-organizing sand dunes
and so causal relationships might be
very much in flux and it puts a higher
focus on action rather than system
description
especially in situations like moving
your head where the relationship of
different things
is going to be heavily action dependent
um and so for now we're kind of in the
systems world where it's like a priori
here the system interfaces
and then we're going to take whether
it's a heuristic or whether you take a
literal interpretation but you don't
have to
you could just say I'm taking
heuristically
the approach of the systems mapping
but that's not to say that that
generative model couldn't be trained in
a systems mapped
context
and then placed into a different setting
where there was a mismatch or a
different generative model
and so it's going to be an ongoing
discussion how are when are these lines
drawn one other point
um other than not needing to use last
names to refer to specific Concepts
um like
are we talking about the blanket that's
statistically insulates every

partition of nodes in these base graphs
which are going to be vast
as Ali and Jacob and I were discussing
on the um branching time active
inference from uh just a few days ago
yesterday
like these graphs are going to be vast
and so just saying there will be a
markout blanket in fact there will be
many Markov blankets with different
semantics that you might assign to them
or not
how does that map onto the real world is
like
the realism instrumentalism
pragmatism abduction conversation that
we've been having do you come in with
these kinds of boundaries
heuristically and distinctions
or how many levels
of uncertainty in nesting I don't know
how many categories of blank there are
I don't know if it's a category or a
Continuum of this
you can actually still take on multiple
levels of uncertainty in the modeling
but then fitting it with empirical data
if it's fit for purpose use is to do
some sort of statistical inference or
like provide meaning or action
but just didactically when the model's
purpose is educational then there's a
lot of things in architectures we might
play with
and then the recipe and also this
conversation uh
helps I hope hold space for the
educational and the Curiosity driven

model development where we're going to
have a lot of fun and interesting
GMS
and
people's applications
and working with different real world
constraints
in different settings we'll all be
learning a lot
from it
and hopefully some of it will be
in repos we can reference
other people may want to play around
inversion within Boulders that we can
make
if anybody wants to be added as a
collaborator in this
textbook repo which is just pretty much
empty
except for I guess a script from Jacob
and there'll be more to say on probably
GitHub and on
um as we continue through but in these
last minutes does anyone have any
thoughts for our continuation of the
cohort
I have one one question uh Daniel you
referenced a paper earlier in the
discussion
um
that I'd love to get the reference to
about uh Persistence of organizations
and survivability
oh yes
um all right
I will put it into
um
into resources

here here's the work

um

it's a PhD dissertation that was
published as a book and I'll upload the
PDF as well this is uh extremely
interesting

work

and okay yeah just it just came out and

um

uh

okay

in our last minutes it's too too fun not
to share

foreign

has written the forward

um

what is a firm's purpose what is an
organization's purpose

it's as difficult to answer as why am I

here free energy governance

and the fep

inverts the question

instead of asking how a system how must
the system behave to survive it asks how
do systems that survive behave

the answer is rather deflationary they

survive their purpose is just

sustainability

so

some classic fristianisms

um some very nice framings on page 95

there's uh

some great juxtapositions

so it's always awesome to see this

kind of quality

work

and

who knows what people here will see and

I don't see any scripts here
that's the textbook but
who knows
maybe we can do some generative modeling
of organizations or we can
specify it and iterate on it
realize it in Block friends and pi mdp
and Forney lab and jf's symbolic sweet
so there'll be a lot of fun things to
explore this cohort
any other questions or ideas that people
want to share
thanks and congratulations to people for
joining this continuation it's a a great
continuance
and
well I know that this part is going to
be very interesting
the next three months will be
interesting